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Solving Stackelberg security Markov games employing the bargaining Nash approach: Convergence analysis

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ABSTRACT

This paper proposes a new approach for the Stackelberg security Markov games' solution, by computing a cooperative approach for the defenders employing the bargaining Nash solution, while the attackers play in a non-cooperative manner. The bargaining Nash solution forces the defenders to negotiate in order to improve their position. A fundamental element of such a game is the disagreement point (status quo), which plays a role of a deterrent. A bargaining solution is defined as a single-valued function that selects an outcome from among the feasible payoffs for each bargaining problem, which in turn is the result of cooperation by the defenders involved in the game. The agreement reached in the game is the most desirable alternative within the set of feasible outcomes. The attackers, playing non-cooperatively, compute the Nash equilibrium point. We employ the Lagrange principle to represent the original game formulation as a nonlinear programming problem. To compute the equilibrium point of the Stackelberg security Markov game, we use an iterative proximal gradient approach. This way, the problem is transformed into a system of equations, which represents an optimization problem for which the necessary condition of a minimum is solved by the projection gradient method. An analysis of the convergence to the Stackelberg security equilibrium point is presented, as well as a random walk solution for planning the patrol schedule which incorporates additional information about the targets using the entropy. The usefulness of the method is successfully demonstrated by a numerical example.

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1. Introduction

1.1. *Brief state-of-the-art review*

The challenge of the Stackelberg security game models is protecting people from delinquent (criminal) attacks and critical infrastructure with limited security resources, which prevent complete security coverage at all times (Trejo et al., 2015b). The research in the area focuses on the dynamics essentially given by defenders that protect a set of targets against attackers. A Stackelberg equilibrium (mixed strategy) is a strategy profile with the property that the defenders first commit to a randomized strategy, to which the attackers devise the best

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response after surveillance of the respective strategy. A player employs a mixed strategy whenever that player chooses to randomize over the set of available actions. A distribution function assigns to each action its probability of being chosen. A mixed strategy profile is a list of strategies, one for each player in the game. A mixed strategy profile induces a probability distribution over the possible outcomes of the game. The dynamics of the game works as follows: the defender first commits to a mixed strategy, and the attacker plays the best response after observing the leader's strategy. Then, the defender considers what the target of the attacker is, and determines his/her strategy, after which, holding the attacked target fixed, the defender picks an action that maximizes his/her payoff.

Game theory provides the mathematical framework for representing the security problem. We outline a Stackelberg game as follows (see also Section 3) (Trejo et al., 2016b). The model can be solved employing a backward induction. Let us consider a game of $\mathcal{N} \cup \mathcal{M}$ players. Let $\mathcal{N}$ denote the set of players called *leaders* (defenders) and let their set of strategies be denoted by U. The rest of $\mathcal{M}$ players are called followers (attackers) and let the set of their strategy profiles be defined by V. Then, $U \times V$ is the complete set of strategy profiles. The dynamics of the game is as follows. The leader considers what the best-reply of the follower is, i.e. how it will respond once it has observed the quantity of the leader. Then, the leaders choose strategy u in U, the followers are informed about u and choose simultaneously their strategies as $\psi(v|u)$ in V, and all players receive their payoffs as given by the strategy profile $(u, \psi(v|u))$. We denote as $\psi(v|u)$ the response of the followers to the strategy u, which is the *best-reply* in the original game (where there is only one follower). In this case, to compute the equilibrium the best-reply of the follower must be calculated first (calculation is backwards because of backward induction). In the security game framework, we suppose that defenders commit to a randomized strategy, while attackers choose their best-reply to this strategy.

A security patrolling example is the Metro. The Metro is a transportation system that connects the metropolitan area of Mexico City by means of underground railways as well as on the surface organized in 12 lines with 195 stations and 226.49 kilometers of route (see Fig. 1). Each line offers one service only, and each line has assigned an identification (letter if feeding line) and a color. Every assigned color is present on square-shaped station logos, system maps and street signs, and neither colors nor numbers have been changed. It serves around 6 million of passengers every day who transfer between lines without additional payment. The security of such a system has a number of challenges. This type of patrol covers the stations of several metro lines. Crimes can be of different types: robbery, assaults, drug dealing, etc. Criminals randomly traverse the metro lines, moving from station to station, seeking opportunities for crime (e.g., riders with mobile-phones). The main goal of the security system is to patrol each of these stations with different security teams like, policemen, canine teams, and civil guards. The game results in a method for coordinating activities between defender's resources.

Stackelberg security games generate random security allocations, like patrols and checkpoints, considering the capacity of the defender resources and the adversary's response. Markov game theory is particularly well-suited for implementing Stackelberg security games and improving the resources allocation using adversarial reasoning. Recent research in security games also proposed the use of random walks to plan patrols under uncertainty. Random walks represent a complementary technique useful for solving scheduling problems (Poznyak et al., 2000). A major advantage of the Markov model is that it is a flexible probabilistic model with modest computational requirements (Markov chains are "memoryless" in the sense that only the current state of the process matters for the future). Moreover, random walks are easy to use, and can easily handle flows around complicated scheduling problems.

1.2. *Main results*

This paper suggests a novel approach for Stackelberg security Markov games, which employs the bargaining Nash solution to compute the cooperation between defenders. Our proposal includes a solution method fixed for Stackelberg security Markov games. The main results are as follows:

The paper presents the following main results:

- Suggests a security game where defenders are engaged cooperatively and the attackers are related non-cooperatively.
- Presents an iterative method for solving Stackelberg security Markov games.
- Transforms the problem in a system of independent equations where each is an optimization problem.
- The method involves two half-steps: the first employs a proximal approach and the second a projection approach.
- Shows the rate of convergence of the method.
- Proposes a random walk solution for planning the patrol schedule, which incorporates additional information about the targets using entropy.
- Presents a numerical example for showing the effectiveness of the method.

1.3. *Organization of the paper*

The rest of the paper is organized as follows. We start with presenting related work in the next section. Section 3 introduces preliminaries about Markov games. Section 4 presents the definitions for Stackelberg games. Bargaining security model is suggested in Section 5. Section 6 discusses the convergence analysis of the Stackelberg security Markov game. A new random walk approach to generate patrols is described in Section 7. Section 8 contains the experimental results showing the usefulness of the method. Section 9 concludes presenting open research issues and future research directions, we also outline the research challenges in this area.

2. Related work

There are several real-world applications to Stackelberg security games (Tambe, 2011). For instance, ARMOR security system



Fig. 1 – Lines of the metro transportation system.

that considers the randomized allocation of checkpoints and canine units at the Los Angeles International Airport (Jain et al., 2010; Pita et al., 2009). IRIS, deployed by the Federal Air Marshals Service, randomizes allocations of air marshals to flights to avoid predictability by adversaries conduction surveillance to provide protection to more important flights (Jain et al., 2011; Tsai et al., 2009). GUARDS (Game-theoretic Unpredictable and Randomly Deployed Security) (Pita et al., 2011) developed by the Transportation Security Administration assists in resource allocation tasks for airport protection focusing in multiple security activities. PROTECT: Port Resilience Operational/Tactical Enforcement to Combat Terrorism under development for the United States Coast Guard (Shieh et al., 2012a; Tambe, 2011) recommends randomized patrolling strategies for the coast guard considering the weights of different targets protected in area of operation. RaPtoR (Varakantham et al., 2013) and TRUSTS (Fave et al., 2014; Yin et al., 2012) generate schedules for patrolling public transit systems (LA Metro and Singapore Metro system). PAWS is a game-theoretic application tested by rangers in Uganda for protecting wildlife

in Queen Elizabeth National Park (Yang et al., 2014). MIDAS was tested by the US Coast Guard for protecting fisheries (Haskell et al., 2014).

Computation of the equilibrium point is a fundamental problem in security games. Korzhyk et al. (2010) devised a method for computing optimal Stackelberg strategies, showing that it can be done in polynomial time in some cases, but is NP-hard in others. Letchford and Conitzer (2013) presented security games where the defender's mission is to monitor the vertices of a graph, and showed how the types of graph, schedules, and defender resources affect the applicability of this approach. They provided a polynomial-time algorithm for computing an optimal defender strategy in one case, and proved NP-hardness of computing an optimal defender strategy in a different case. Xu (2016) described security games from a theoretical point of view, suggesting that each security game can be described by a set system, and showing that the computational complexity of a security game is essentially determined by this set system.

Kiekintveld et al. (2009) suggested a compact representation technique, in which security games can be described by a polynomial-sized mixed-integer linear programming problem. Korzhyk et al. (2010) used the aforementioned compact representation of such games (Kiekintveld et al., 2009) to compute optimal Stackelberg strategies, and showed that this can be done in polynomial time in some cases, while it is NP-hard in others. Korzhyk et al. (2011), based on an idea of von Stengel and Zamir (2010), performed a transformation of a Stackelberg game into a Nash game, relaxing the interpretation of the solution and the complexity of the equilibrium point selection problem. Xu et al. (2014) suggested an algorithm for computing minimax strategies when the problem admits a polynomial-time oracle, showing that the spatial and temporal security game is NP-hard in the general case. Bhattacharya et al. (2011) considered a model which minimizes the total cost of the purchased resources, given that every target has to be defended with at least a certain probability, and proved that the general security game with costly resources is NP-hard. There are other lines of research related to security games, for instance the Bayesian security game (Kiekintveld et al., 2011; Paruchuri et al., 2008), the quantal response (Shieh et al., 2012b), or the uncertain attacker behavior (Balcan et al., 2015). The aforementioned papers focus on the security game with a single defender and attacker, and the computational complexity thereof. None of them provides a solution for the security game considering multiple defenders and attackers.

By casting the problem as a Stackelberg game for multiple players in the context of Markov chain games, we conceptualized security games as polylinear games, and we have developed new algorithms and methods for solving security games, which provide randomized patrolling strategies for optimizing resource allocation. We introduced specific techniques to handle each of these challenges: non-cooperative, cooperative, Lyapunov approach, and ultimately negotiation by bargaining. These solutions have led to successes and advances using random walk techniques in security scheduling and planning. Trejo et al. (2015b) presented an approach for representing a real-world attacker-defender Stackelberg security game-theoretical model based on Markov chain games, and employed the Lagrange principle and Tikhonov's regularization method to ensure the convergence of the cost functions to the respective equilibria. They based their results on a previous paper (Trejo et al., 2015a), developing the extraproximal method for computing the Stackelberg/Nash equilibria. Clempner and Poznyak (2015) considered a game-theory approach for representing a real-world attacker-defender Stackelberg security game where the behavior of an ergodic system is represented by a Lyapunov-like function that is non-decreasing in time. Then, the representation of the Stackelberg security game is transformed into a potential game in terms of Lyapunov. Clempner and Poznyak (2016b) suggested a security model that describes a strategic game in which the defenders cooperate and attackers do not cooperate. Trejo et al. (2016a) suggested a cooperative approach for adapting strategies in Stackelberg security Markov games using a Reinforcement Learning (RL) method, where attackers and defenders employ average rewards. The proposed framework combines prior knowledge and temporal-difference methods included in the RL architecture, involving two highest components: the adaptive primary learning architecture and the actor-critic architecture. In the case of a metric state space, the coalition of the defenders achieves it synergy by computing the Strong L_p–Stackelberg/Nash equilibrium (Trejo et al., 2016b, 2017). Clempner and Poznyak (2017) extended the model presented in Clempner and Poznyak (2015), by solving the security game problem using the extraproximal approach for computing the shortest-path Lyapunov equilibrium in Stackelberg security games. The extraproximal method is employed to compute the mixed stationary strategies: attackers operate on partial knowledge of the defenders' strategies for fixed targets. Clempner (2017) provided an approach for handling real-world patrolling domains with spatiotemporal constraints based on continuous-time Stackelberg security Markov games. Guerrero et al. (2017) proposed an iterative method for computing the equilibrium point in Stackelberg security Markov games casting the problem for multiple players in polylinear games.

3. Markov game

Let the set of all nonnegative integers be denoted by $\mathbb{N}$. A Borel subset X of a complete and separable metric space is called a Borel space, and its Borel σ-algebra is denoted by $\mathbb{B}(X)$. Let X and Y be Borel spaces. A *stochastic kernel* on X given Y is a function $P(\cdot|\cdot)$ such that $P(\cdot|y)$ is a probability measure on X for every fixed $y \in Y$, and $P(B|\cdot)$ is a measurable function on Y for every fixed $B \in \mathbb{B}(X)$. If $X = Y$, then P is called a (Markov) *transition kernel*.

Let $M = (S, A, \{A(s)\}_{s\in S}, P)$ be a Markov chain (Clempner and Poznyak, 2014; Poznyak et al., 2000), where S is a finite set of *states*, $S \subset \mathbb{N}$ and A is a finite *set of actions*. For each $s \in S$, $A(s) \subset A$ is the non-empty set of admissible actions at state $s \in S$. Without loss of generality we may take $A = \cup_{s\in S} A(s)$. Whereas, $\mathbb{K} = \{(s,a)|s \in S, a \in A(s)\}$ is the set of *admissible state-action pairs*, which is a measurable subset of $S \times A$. The variable p_{jlik} is a stationary controlled *transition matrix*, which defines a stochastic kernel on S given $\mathbb{K}$, where

$p_{j|ik} := P(X_{t+1} = s_j | X_t = s_i, A_t = a_k)$ $\forall t \in \mathbb{N}$ represents the probability associated with the transition from state s_i to state s_j, $i = \overline{1,N}$ $(i = 1,\dots,N)$ and $j = \overline{1,N}$ $(j = 1,\dots,N)$, under an action $a_k \in A(s_i)$, $k = \overline{1,K}$ $(k = 1,\dots,K)$. A *Markov Decision Process* is a pair $MDP = (M, C)$ where M is a controllable Markov chain and $C : \mathbb{K} \to \mathbb{R}$ is a *cost function*, associating to each state-action pair a real value.

This *MDP* is interpreted as follows: At each time $t \in \mathbb{N}$ the decision maker observes the state of a dynamical system, say $X_t = s_i \in S$, and selects the action (control) $A_t = a_k \in A(s_i)$. Then, a cost $C_{(s,a)}$ is incurred and, regardless of the previous states and actions, the state of the system at time $t+1$ will be $X_{t+1} = s_j \in S$ with probability $p_{j|ik}$; this is the Markov property of the decision process. Let $\mathbb{F}$ denote the set of measurable functions $f: S \to A$ such that $f(s)$ is in $A(s)$ for all $s \in S$, and Φ stands for the set of stochastic kernels φ on A given S for which $\varphi(A(s)|s) = 1$ for all $s \in S$.

The space $\mathbb{H}_t$ of possible histories up to time $t \in \mathbb{N}$ is defined by $\mathbb{H}_0 := S$ and $\mathbb{H}_t := \mathbb{K}_t \times S$, $t \geq 1$, $h_t = (s_0, a_0, \dots, s_i, a_i, \dots, s_t)$ stands for a generic element of $\mathbb{H}_t$, where $a_k \in A(s_i)$. A *control policy* $\pi = \{\pi(t)\}$ is a special sequence of stochastic kernels: for each $t \in \mathbb{N}$ and $h_t \in \mathbb{H}_t$, $\pi_{(\cdot|h_t)}(t)$ is a probability measure on A concentrated on $A(s_t)$ then the controller chooses actions according to the control A_t applied at time t belonging to A with probability $\pi_{(A|h_t)}(t) = 1$, where h_t is the observed history of the process up to time t. Then, the policy $\pi_{(k|i)}(t) := P(A_t = a_k | X_t = s_i)$ represents the probability measure associated with the occurrence of an action $A_t = a_k$ from state $X_t = s_i$ at time $t \in \mathbb{N}$.

Let Π be the class of all policies. Given the π policy being used for choosing actions and the initial state $X_0 = s \in S$, the distribution of the state-action process denoted by $\{(S_t, A_t) | t \in \mathbb{N}\}$ is uniquely determined and such a distribution and the corresponding expectation operator are P and E respectively.

A game consists of a set $\mathcal{N} = \{1,\dots,n\}$ of players (indexed by $l = \overline{1,n}$). The dynamics of a Markov game is described as follows. Each of the players l is allowed to randomize actions, with probabilities $\pi^l_{k|i}(t)$, over the pure action choices $a^l_k \in A^l(s^l_i)$, $i = \overline{1,N}$ and $k = \overline{1,K}$. Herein, we will consider only stationary strategies $\pi^l_{k|i}(t) = \pi^l_{k|i}$. In the ergodic case when all Markov chains are ergodic for any stationary strategy $\pi^l_{k|i}$ the probabilities $P^l(X^l_{t+1} = s^l_j)$ exponentially quickly converge to their limit probabilities, given by the solution to the linear system of equations $P^l(s^l_j) = \sum_{i=1}^{N} \left(P^l(s_i) \sum_{k=1}^{K} p^l_{j|ik} \pi^l_{k|i} \right)$.

The expected cost function J^l of each player l depends on the states and actions of all the other players and it is given by the values of $V^l_{i,k}$ as follows

$$J^l(z^1, .., z^n) := E\left\{ \sum_i^N \sum_k^K V^l_{ik} \prod_{l=1}^n z^l_{ik} \right\} \tag{1}$$

given that

$$V^l_{ik} = \sum_i^N \sum_k^K \left(\sum_j^N C^l_{ik} \prod_{l=1}^n p^l_{j|ik} \right) \tag{2}$$

and $z^l := [z^l_{ik}]_{i=\overline{1,N}; k=\overline{1,K}}$ $(l = \overline{1,n})$ is a matrix with elements

$$z^l_{ik} = \pi^l_{k|i} P^l(s^l_i) \tag{3}$$

satisfying

$$z^l \in Z^l_{adm} = \begin{cases} z^l : \sum_i^N \sum_k^K z^l_{ik} = 1, z^l_{ik} \geq 0 \\ h^l_j(z) = \\ \sum_i^N \sum_k^K p^l_{j|ik} z^l_{ik} - \sum_k^K z^l_{jk} = 0 \end{cases} \tag{4}$$

Note that by (3) it follows that

$$P^l(s^l_i) = \sum_k^K z^l_{ik} \qquad \pi^l_{k|i} = \frac{z^l_{ik}}{\sum_k^K z^l_{ik}} \tag{5}$$

In the ergodic case $\sum_k^K z^l_{ik} > 0$ for all $l = \overline{1,n}$. The *individual aim* of each player is

$$J^l(z) \to \min_{z \in Z_{adm}}.$$

4. Stackelberg security game

4.1. *Stackelberg game*

Let us consider a game (Clempner and Poznyak, 2016a; Solis et al., 2016; Trejo et al., 2015a) *with a set* $\mathcal{N}$ *of defenders* $(l = \overline{1,n})$ whose strategies are denoted by $u^l \in U^l$ where U is a convex and compact set where

$$u^l := col(z^l_{ik}),$$
$$U^l := Z^l_{adm}\left(l = \overline{1,n}\right), U := \bigotimes_{l=1}^{n} U^l$$

where *col* is the column operator which transforms the matrix $z^l_{(ik)}$ into a column. Denoted by $u = (u^1, \dots, u^n)^{\mathrm{T}} \in U$, the joint strategy of the players, and $\hat{u} = u^{-l}$ is a strategy of the rest of the players adjoint to u^l, namely,

$$u^{-l} := (u^1, \dots, u^{l-1}, u^{l+1}, \dots, u^n)^{\mathrm{T}} \in U^{-l} := \bigotimes_{h=1, h \neq l}^{n} U^h$$

such that $u = \left(u^l, u^{\hat{l}}\right)$.

Similarly, let us consider *a set* $\mathcal{M}$ of attackers with strategies $v^r \in V^r$ $\left(r = \overline{1,m}\right)$ and V is also a convex and compact set such that

$$v^r := col(z^r_{ik}), V^r := Z^r_{adm}\left(r = \overline{1,m}\right), V := \bigotimes_{r=1}^{m} V^r$$

Denoted by $v = (v^1, \dots, v^m) \in V := \bigotimes_{r=1}^{m} V^r$ the joint strategy of the attackers and $v^{\hat{r}} = v^{-r}$ is a strategy of the rest of the players adjoint to v^r, namely,

$$v^{-r} := (v^1, \dots, v^{m-1}, v^{m+1}, \dots, v^m)^{\mathrm{T}} \in V^{-r} := \bigotimes_{r=1, r \neq m}^{m} V^r$$

such that $v = \left(v^r, v^{\hat{r}}\right)$ $\left(r = \overline{1,m}\right)$.

The dynamics of the game is as follows. The *defenders play non-cooperatively* and they are assumed to anticipate the reactions of the attackers trying to reach the Nash equilibria. For reaching the goal of the game *defenders* first try to find a joint strategy $u^* = (u^{1*}, \dots, u^{n*}) \in U$ satisfying for any admissible $u^l \in U^l$ and any $l = \overline{1,n}$

$$G(u) := \sum_{l=1}^{n} \left[\left(\min_{u^l \in U^l} \vartheta_l(u^l, u^{-l}) \right) - \vartheta_l(u^l, u^{-l}) \right] \tag{6}$$

(Tanaka, 1989; Tanaka and Yokoyama, 1991). Here $\varphi_l(u^l, u^{-l})$ is the cost function of the leader l which plays the strategy $u^l \in U^l$ and the rest of the leaders play the strategy $u^{-l} \in U^{-l}$.

If we consider the utopia point

$$\bar{u}^l := \arg\min_{u^l \in U^l} \varphi_l(u^l, u^{-l}) \tag{7}$$

then, we can rewrite Eq. (6) as follows

$$G(u) := \sum_{l=1}^{n} [\varphi_l(\bar{u}^l, u^{-l}) - \varphi_l(u^l, u^{-l})]. \tag{8}$$

The functions $\varphi_l(u^l, u^{-l})$ and $(l = \overline{1,n})$ are assumed to be convex in all their arguments.

The function $G(u)$ satisfies *the* **Nash condition**

$$\max_{u \in U} g(u) = \sum_{l=1}^{n} [\varphi_l(\bar{u}^l, u^{-l}) - \varphi_l(u^l, u^{-l})] \le 0$$

for any $u^l \in U^l$ and all $l = \overline{1,n}$.

A strategy $u^* \in U_{adm}$ is said to be a **Nash equilibrium** if

$$u^* \in Arg\min_{u \in U_{adm}} \{G(u)\}.$$

If $G(u)$ is strictly convex then

$$u^* = \arg\min_{u \in U_{adm}} \{G(u)\}.$$

As well, in this process the attackers try to reach one of the Nash equilibria trying to find a joint strategy $v^* = (v^{1*}, \dots, v^{m*}) \in V$ satisfying for any admissible $v^r \in V^r$ and any $r = \overline{1,m}$

$$F(v) := \sum_{r=1}^{m} \left[\left(\min_{v^r \in V^r} \psi_r(v^r, v^{-r}) \right) - \psi_r(v^r, v^{-r}) \right]$$

(Tanaka, 1989; Tanaka and Yokoyama, 1991). Here $\psi_r(v^r, v^{-r})$ is the cost function of the follower m which plays the strategy $v^r \in V^r$ and the rest of the leaders play the strategy $v^{-r} \in V^{-r}$.

If we consider the utopia point

$$\bar{v}^r := \arg\min_{v^r \in V^r} \psi_r(v^r, v^{-r})$$

then, we can rewrite Eq. (6) as follows

$$F(v) := \sum_{r=1}^{m} (\psi_r(\bar{v}^r, v^{-r}) - \psi_r(v^r, v^{-r})).$$

The functions $\psi_r(v^r, v^{-r})$ $(r = \overline{1,m})$ are assumed to be convex in all their arguments.

The function $F(v)$ satisfies *the* **Nash condition**

$$\max_{v^r \in V^r} f(v) = \sum_{r=1}^{m} (\psi_r(\bar{v}^r, v^{-r}) - \psi_r(v^r, v^{-r})) \le 0$$

for any $v^r \in V^r$ and all $r = \overline{1,m}$.

Defenders and *attackers* together are in a Stackelberg game: the model involves two non-cooperatively Nash games restricted by a Stackelberg game defined as follows.

Definition 1. *A game with* n *defenders and* m *attackers is said to be a Stackelberg–Nash game if*

$$G(u|v) := \sum_{l=1}^{n} (\varphi_l(\bar{u}^l, u^{-l}|v) - \varphi_l(u^l, u^{-l}|v))$$

such that

$$\max_{u \in U} g(u|v) = \sum_{l=1}^{n} [\varphi_l(\bar{u}^l, u^{-l}|v) - \varphi_l(u^l, u^{-l}|v)] \le 0$$

where u^{-l} *is a strategy of the rest of the defenders adjoint to* u^l*, namely,*

$$u^{-l} := (u^1, \dots, u^{l-1}, u^{l+1}, \dots, u^n) \in U^{-l} := \bigotimes_{h=1, h \ne l}^{n} U^h$$

and

$$\bar{u}^l := \arg\min_{u^l \in U^l} \varphi_l(u^l, u^{-l}|v)$$

such that

$$f(v|u) := \sum_{r=1}^{m} (\psi_r(\bar{v}^r, v^{-r}|u) - \psi_r(v^r, v^{-r}|u))$$

given that v^{-r} *is a strategy of the rest of the attackers adjoint to* v^r*, namely,*

$$v^{-r} := (v^1, \dots, v^{m-1}, v^{m+1}, \dots, v^m) \in V^{-r} := \bigotimes_{q=1, q \ne l}^{m} V^q$$

and

$$\bar{v}^r := \arg\min_{v^r \in V^r} \psi_r(v^r, v^{-r}|u).$$

4.2. *Lagrange regularization approach*

Let us consider the regularized Lagrange function given by

$$\mathbb{L}_{\theta,\delta}(u, v, \lambda) := \theta G(u|v) + \lambda^{\mathsf{T}} (A^u_{eq} u - b_{eq}) + \lambda^{\mathsf{T}} (A^v_{eq} v - b_{eq}) + \frac{\delta}{2} \left(\|u\|^2 - \|v\|^2 - \|\lambda\|^2 \right) \tag{9}$$

where the parameters θ, δ are positive and the Lagrange vector-multipliers and the components of $\lambda \in \Lambda \subseteq \mathbb{R}$ may have any sign. Then, the optimization problem

$$\mathbb{L}_{\theta,\delta}(u,v,\lambda)\to\min_{u\in U_{adm}}\max_{v\in V_{adm}}\max_{\lambda} \tag{10}$$

has a unique saddle point on u, v and λ since the optimized regularized Lagrange function given in Eq. (9) is *strongly convex* (Poznyak, 2008) if the parameters θ and $\delta>0$ provide the condition

$$\frac{\partial^2}{\partial x\partial x^{\mathrm{T}}}\mathbb{L}_{\theta,\delta}(u,v,\lambda)>0\forall x^*\in U^*\otimes V^*\otimes\Lambda^* \tag{11}$$

and it is strongly concave on v and the Lagrange multiplier λ for any $\delta>0$. The regularized Lagrange function has a unique saddle point $(u^*(\theta,\delta),v^*(\theta,\delta),\lambda^*(\theta,\delta))$ because

$$\begin{aligned}&\mathbb{L}_{\theta,\delta}(u(\theta,\delta),v^*(\theta,\delta),\lambda^*(\theta,\delta))\geq\\&\mathbb{L}_{\theta,\delta}(u^*(\theta,\delta),v^*(\theta,\delta),\lambda^*(\theta,\delta))\geq\\&\mathbb{L}_{\theta,\delta}(u^*(\theta,\delta),v(\theta,\delta),\lambda(\theta,\delta))\end{aligned} \tag{12}$$

For the non-regularized Lagrange function given by $\mathbb{L}_{1,0}(u,v,\lambda)$, it may have several (not obligatory unique) saddle points $(u^*,v^*,\lambda^*)\in U^*\otimes V^*\otimes\Lambda^*$.

If the parameter θ and the regularizing parameter δ tend to zero by a particular manner, then we may expect that $u^*(\theta,\delta)$, $v^*(\theta,\delta)$ and $\lambda^*(\theta,\delta)$ which are the solutions of the min-max optimization problem (9) tend to the set $U^*\otimes V^*\otimes\Lambda^*$ of all saddle points of the original optimization problem, that is,

$$d\{u^*(\theta,\delta),v^*(\theta,\delta),\lambda^*(\theta,\delta);U^*\otimes V^*\otimes\Lambda^*\}\underset{\theta,\delta\downarrow 0}{\to}0 \tag{13}$$

where $d\{a;U^*\otimes V^*\otimes\Lambda^*\}$ is the Hausdorff distance defined as

$$d\{a;U^*\otimes V^*\otimes\Lambda^*\}=\min_{x^*\in U^*\otimes V^*\otimes\Lambda^*}\|a-x^*\|^2$$

Finally, the regularized Lagrange function for the Stackelberg security game is given by

$$\begin{aligned}\mathbb{L}_{\theta,\delta}(u^l,u^{-l},v^r,v^{-r},\omega,\xi,\lambda):=&\underbrace{\theta G(u|v)-\omega g(u|v)}_{(\theta-\omega)G(u|v)}+\\&\xi f(v|u)+\lambda^{\mathrm{T}}\left(A^u_{eq}u-b_{eq}\right)+\lambda^{\mathrm{T}}\left(A^v_{eq}v-b_{eq}\right)+\\&\tfrac{\delta}{2}\left(\|u^l\|^2+\|u^{-l}\|^2-\|v^r\|^2-\|v^{-r}\|^2+\|\omega\|^2-\|\xi\|^2-\|\lambda\|^2\right)\end{aligned}$$

where for Markov games

$$A^u_{eq}=\begin{bmatrix}h^l_j(u)=\left([p^{l,j}]\right)^{\mathrm{T}}u^l-u^{l,j}\\u^le=1,u^l\geq 0\end{bmatrix}$$

and $p^{l,j}=col\ p^{(l,j)}_{(i|k)}$ (col is the column operator), $u^{l,j}=col\ z^l_{jk}$, and $e=(1,\dots,1)^{\mathrm{T}}\in\mathbb{R}^{(n)}$. We have a similar development for A^v_{eq}.

5. Bargaining security model

5.1. Bargaining game

For a set $\mathcal{N}$ *of defenders* $\left(l=\overline{1,n}\right)$ whose strategies are denoted by $u^l\in U^l$ where U is a convex and compact set, a Nash bargaining game (Trejo and Clempner, 2018) is defined by a pair $(\Delta,\mathcal{B})$, where $\mathcal{B}\subseteq\mathbb{R}^n_+$ is a compact and convex set which defines the feasible set of utilities of all the defenders, and Δ is known as the disagreement point (status quo) which plays the role of a deterrent.

Nash (1950) defined the bargaining strategy solution $u^*\in U$ of this game to be the one which satisfies four axioms: 1) Pareto optimality, 2) Invariance under transformation of utilities, 3) Symmetry and, 4) Independence of irrelevant alternatives. Nash (1950) proved that there is a unique strategy u^* such that $\varphi(u^{l*},u^{-l*})\in\mathcal{B}$ which satisfies these axioms, and moreover this point $(\varphi(u^{l*},u^{-l*})\in\mathcal{B})$ is the one that maximizes $\prod_{l=1}^n(\varphi(u^l,u^{-l})-\Delta^l)$ or equivalently $\sum_{l=1}^n\ln(\varphi(u^l,u^{-l})-\Delta^l)$.

In the nonsymmetric bargaining game, each player l has a positive bargaining weight α_l which represents the relative bargaining power. The bargaining solution in this case is the point in the feasible region $\mathcal{B}$ which maximizes the objective function $\sum_{l=1}^n\ln(\varphi(u^l,u^{-l})-\Delta^l)$.

5.2. Stackelberg security bargaining game

Defenders and attackers together are in a Stackelberg game: the model involves a cooperative bargaining Nash solution (Trejo and Clempner, 2018) for the defenders and a non-cooperative Nash game for the attackers restricted by a Stackelberg game which is defined as follows.

Definition 2. *A game with n defenders and m attackers is said to be a Stackelberg–Nash security game if*

$$G(u|v):=\prod_{l=1}^n\left(\varphi^l(u^l,u^{-l}|v)-\Delta^l\right)^{\alpha^l\chi(\varphi^l>\Delta^l)}=\min_{u\in U}g(u|v)$$

such that

$$\min_{u\in U}g(u|v)=$$
$$-\max_{u\in U}\sum_{l=1}^n\alpha^l\chi\left(\varphi^l(u^l,u^{-l}|v)>\Delta^l\right)\ln\left(\varphi^l(u^l,u^{-l}|v)-\Delta^l\right)$$

where $\hat{u}=u^{-l}$ *is a strategy of the rest of the defenders adjoint to* u^l*, namely,*

$$u^{-l}:=\left(u^1,\dots,u^{l-1},u^{l+1},\dots,u^n\right)\in U^{-l}:=\bigotimes_{h=1,h\neq l}^n U^h$$

such that

$$f(v|u):=\sum_{r=1}^m\left(\psi_r(\bar{v}^r,v^{-r}|u)-\psi_r(v^r,v^{-r}|u)\right) \tag{14}$$

given that $\hat{v}=v^{-r}$ *is a strategy of the rest of the attackers adjoint to* v^r*, namely,*

$$v^{-r}:=\left(v^1,\dots,v^{m-1},v^{m+1},\dots,v^m\right)\in V^{-r}:=\bigotimes_{q=1,q\neq l}^m V^q$$

and

$$\bar{v}^r:=\arg\min_{v^r\in V^r}\psi_r(v^r,v^{-r}|u).$$

Remark 3. *We employ the restriction f(v|u) for ensuring the attackers to play non-cooperatively.*

Definition 4. *Let G(u|v) be the cost functions of the defenders* $(l=\overline{1,n})$. *A strategy* $u^* \in U$ *of the defenders together with the collection* $v^* \in V$ *of the attackers is said to be* ***a Stackelberg–Nash equilibrium*** *if*

$$(u^*, v^*) \in Arg\min_{u\in U}\min_{\hat{u}(u)\in\hat{U}}\max_{v\in V,}\max_{\hat{v}(v)\in\hat{V}} \{G(u|v)|,\ f(v|u)\le \mathbf{0}\}. \tag{15}$$

Remark 5. *If G(u|v) is strictly convex then*

$$(u^*, v^*) = \arg\min_{u\in U}\min_{\hat{u}(u)\in\hat{U}}\max_{v\in V}\max_{\hat{v}(v)\in\hat{V}} \{G(u|v)|f(v|u)\le 0\}$$

5.3. *Regularization for the bargaining game*

Let us consider a bargaining game for a set $\mathcal{N}$ *of defenders* $(l=\overline{1,n})$. Let us denote the disagreement utility that depends on the strategies z_{ik}^l as $\varphi^{l\Delta}(u^l, u^{-l}|v)$ for each defender $(l=\overline{1,n})$, and the solution for the Nash bargaining problem as the point $(\varphi^1,\dots,\varphi^n)$. The utilities φ^l, in the same way that the disagreement utilities, are for Markov chains as follows

$$\varphi^l = \varphi^l(u^l, u^{-l}|v) := \sum_i^N \sum_k^K V_{ik}^l \prod_{l=1}^{n+m} z_{ik}^l \tag{16}$$

where the utility V^l is defined in Eq. (2). This point is better than the disagreement point, therefore it must satisfy that $\varphi^l > \varphi^{l\Delta}$.

The process of solving the bargaining problem consists of two main steps: 1) to find the disagreement point (in our case we define the disagreement point as the Nash equilibrium point of the problem (Nash, 1951)) and; 2) for the solution of the bargaining process we follow the model presented by Nash (1950).

The function for finding the solution to the Nash bargaining problem for the defenders is given by

$$g(u^l, u^{-l}|v) = \prod_{l=1}^{n} \left(\varphi^l(u^l, u^{-l}|v) - \Delta^l\right)^{\alpha^l \chi(\varphi^l > \Delta^l)}$$

where $\alpha^l \ge 0$ and $\sum_{l=1}^n \alpha^l = 1$, which are weighting parameters for each defender. We can rewrite the previous equation as follows

$$\tilde{g}(u^l, u^{-l}|v) = \sum_{l=1}^{n} \alpha^l \chi(\varphi^l(u^l, u^{-l}|v) > \Delta^l) \ln\left(\varphi^l(u^l, u^{-l}|v) - \Delta^l\right)$$

Thus, the strategy u^*, which is the vector $u^* = (u^1,\dots,u^n) \in U_{adm} := \otimes_{l=1}^n Z_{adm}^l$, is the solution for the Nash bargaining problem

$$u^* \in arg\max_{u\in U_{adm}} \{\tilde{g}(u^l, u^{-l}|v)\}$$

where the strategies u^l satisfy the restrictions in Eq. (4).

Applying the Lagrange principle and the Tikhonov's regularization for the Stackelberg security bargaining game we have that

$$\begin{aligned}\mathbb{L}_{\theta,\delta}(u^l, u^{-l}, v^r, v^{-r}, \xi, \lambda) :=\\ \theta G(u|v) + \xi f(v|u) +\\ \lambda^T\left(A_{eq}^u u - b_{eq}\right) + \lambda^T\left(A_{eq}^v v - b_{eq}\right) +\\ \frac{\delta}{2}\left(\|u^l\|^2 + \|u^{-l}\|^2 - \|v^r\|^2 -\right.\\ \left.\|v^{-r}\|^2 - \|\xi\|^2 - \|\lambda\|^2\right)\end{aligned}$$

6. Solver of the Stackelberg security Markov game

6.1. *Iterative method*

Note that the function $\mathbb{L}_{\theta\delta}(u, \hat{u}, v, \hat{v}, \xi, \lambda)$ is polylinear in $u \in U_{adm}, \hat{u} \in \hat{U}_{adm}, v \in V_{adm}$, $\hat{v} \in \hat{V}_{adm}$, and $\xi, \lambda \in \Lambda$, and therefore can not be solved analytically. Then, we need to apply an iterative method to find a minimizing solution which, additionally, may be not unique. The general-format iterative version $(n = 0, 1, \dots)$ of the iterative step method for computing the Stackelberg security equilibrium point with some fixed admissible initial values $(u_0 \in U,\ \hat{u}_0(u) \in \hat{U}, v_0 \in V,\ \hat{v}_0(v) \in \hat{V}, \xi_0, \lambda_0 \in \Lambda)$ is as follows

1. Proximal prediction step:

$$\begin{aligned}\bar{\xi}_n &= \arg\min_{\xi\ge 0}\left\{\frac{1}{2}\|\xi - \xi_n\|^2 - \gamma\mathbb{L}_{\theta\delta}(u_n, \hat{u}_n, v_n, \hat{v}_n, \xi, \bar{\lambda}_n)\right\}\\ \bar{\lambda}_n &= \arg\min_{\lambda\ge 0}\left\{\frac{1}{2}\|\lambda - \lambda_n\|^2 - \gamma\mathbb{L}_{\theta\delta}(u_n, \hat{u}_n, v_n, \hat{v}_n, \bar{\xi}_n, \lambda)\right\}\\ \bar{u}_n &= \arg\min_{u\in U_{adm}}\left\{\frac{1}{2}\|u - u_n\|^2 + \gamma\mathbb{L}_{\theta\delta}(u, \hat{u}_n, v_n, \hat{v}_n, \bar{\xi}_n, \bar{\lambda}_n)\right\}\\ \bar{\hat{u}}_n(u) &= \arg\min_{\hat{u}\in\hat{U}_{adm}}\left\{\frac{1}{2}\|\hat{u} - \hat{u}_n\|^2 + \gamma\mathbb{L}_{\theta\delta}(u_n, \hat{u}, v_n, \hat{v}_n, \bar{\xi}_n, \bar{\lambda}_n)\right\}\\ \bar{v}_n &= \arg\min_{v\in V_{adm}}\left\{\frac{1}{2}\|v - v_n\|^2 - \gamma\mathbb{L}_{\theta\delta}(u_n, \hat{u}_n, v, \hat{v}_n, \bar{\xi}_n, \bar{\lambda}_n)\right\}\\ \bar{\hat{v}}_n(v) &= \arg\min_{\hat{v}\in\hat{V}_{adm}}\left\{\frac{1}{2}\|\hat{v} - \hat{v}_n\|^2 - \gamma\mathbb{L}_{\theta\delta}(u_n, \hat{u}_n, v_n, \hat{v}, \bar{\xi}_n, \bar{\lambda}_n)\right\}\end{aligned} \tag{17}$$

2. Gradient aproximation step:

$$\begin{aligned}\xi_{n+1} &= \xi_n + \gamma\nabla_{\xi}\mathbb{L}_{\theta\delta}(\bar{u}_n, \bar{\hat{u}}_n, \bar{v}_n, \bar{\hat{v}}_n, \xi, \bar{\lambda}_n)\\ \lambda_{n+1} &= \lambda_n + \gamma\nabla_{\lambda}\mathbb{L}_{\theta\delta}(\bar{u}_n, \bar{\hat{u}}_n, \bar{v}_n, \bar{\hat{v}}_n, \bar{\xi}_n, \lambda)\\ u_{n+1} &= \Pr_{u\in U_{adm}}\left\{u_n - \gamma\nabla_u\mathbb{L}_{\theta\delta}(u, \bar{\hat{u}}_n, \bar{v}_n, \bar{\hat{v}}_n, \bar{\xi}_n, \bar{\lambda}_n)\right\}\\ \hat{u}_{n+1}(u) &= \Pr_{\hat{u}\in\hat{U}_{adm}}\left\{\hat{u}_n(u) - \gamma\nabla_{\hat{u}}\mathbb{L}_{\theta\delta}(\bar{u}_n, \hat{u}, \bar{v}_n, \bar{\hat{v}}_n, \bar{\xi}_n, \bar{\lambda}_n)\right\}\\ v_{n+1} &= \Pr_{v\in V_{adm}}\left\{v_n + \gamma\nabla_v\mathbb{L}_{\theta\delta}(\bar{u}_n, \bar{\hat{u}}_n, v, \bar{\hat{v}}_n, \bar{\xi}_n, \bar{\lambda}_n)\right\}\\ \hat{v}_{n+1}(v) &= \Pr_{\hat{v}\in\hat{V}_{adm}}\left\{\hat{v}_n + \gamma\nabla_{\hat{v}}\mathbb{L}_{\theta\delta}(\bar{u}_n, \bar{\hat{u}}_n, \bar{v}_n, \hat{v}, \bar{\xi}_n, \bar{\lambda}_n)\right\}\end{aligned} \tag{18}$$

6.2. *Convergence of the Stackelberg security game*

Let us define the following extended vectors

$$\tilde{u} := \begin{pmatrix} u \\ \hat{u} \end{pmatrix} \in \tilde{U} := U \times \hat{U},$$

$$\tilde{v} := \begin{pmatrix} v \\ \hat{v} \\ \xi \\ \lambda \end{pmatrix} \in \tilde{V} := V \times \hat{V} \times \mathbb{R}^+ \times \mathbb{R}^+.$$

Then, the regularized Lagrange function can be expressed as

$$\tilde{\mathbb{L}}_{\theta\delta}(\tilde{u}, \tilde{v}) := \mathbb{L}_{\theta\delta}(u_\delta, \hat{u}_\delta, v_\delta, \hat{v}_\delta, \xi_\delta, \lambda_\delta).$$

The equilibrium point that satisfies Eqs. 17 and Eqs. 18 can be expressed as

$$\tilde{u}^*_\delta = \mathrm{Pr}_{\tilde{u}\in\tilde{U}_{adm}}\{\tilde{u} - \gamma\nabla_{\tilde{u}}\tilde{\mathbb{L}}_{\theta\delta}(\tilde{u}, \tilde{v}^*_\delta)\}$$
$$\tilde{v}^*_\delta = \mathrm{Pr}_{\tilde{v}\in\tilde{V}_{adm}}\{\tilde{v} - \gamma\nabla_{\tilde{v}}\tilde{\mathbb{L}}_{\theta\delta}(\tilde{u}^*_\delta, \tilde{v})\}.$$

Now let us introduce the following variables

$$\tilde{x} = \begin{pmatrix} \tilde{x}_1 \\ \tilde{x}_2 \end{pmatrix} \in \tilde{X} := \tilde{U} \times \tilde{V}$$
$$\tilde{y} = \begin{pmatrix} \tilde{y}_1 \\ \tilde{y}_2 \end{pmatrix} \in \tilde{Y} := \tilde{U} \times \tilde{V}$$

and define the Lagrangian in terms of the previous variables

$$L_{\theta\delta}(\tilde{x}, \tilde{y}) := \tilde{\mathbb{L}}_{\theta\delta}(\tilde{x}_1, \tilde{y}_2) - \tilde{\mathbb{L}}_{\theta\delta}(\tilde{y}_1, \tilde{x}_2).$$

For $\tilde{x}_1 = \tilde{u}$, $\tilde{x}_2 = \tilde{v}$, $\tilde{y}_1 = \tilde{y}^*_1 = \tilde{u}^*_\delta$ and $\tilde{y}_2 = \tilde{y}^*_2 = \tilde{v}^*_\delta$ we have

$$L_{\theta\delta}(\tilde{x}, \tilde{y}^*) := \tilde{\mathbb{L}}_{\theta\delta}(\tilde{u}, \tilde{v}^*_\delta) - \tilde{\mathbb{L}}_{\theta\delta}(\tilde{u}^*_\delta, \tilde{v}).$$

In these variables the proximal format can be represented by

$$\tilde{y}^* = \arg\min_{\tilde{x}\in\tilde{U}\times\tilde{V}}\left\{\tfrac{1}{2}\|\tilde{x} - \tilde{y}^*\|^2 + \gamma L_{\theta\delta}(\tilde{x}, \tilde{y}^*)\right\}. \tag{19}$$

Finally, we have that the iterative method can be expressed by

1. First step

$$\bar{\tilde{y}}_n = \arg\min_{\tilde{x}\in\tilde{U}\times\tilde{V}}\left\{\tfrac{1}{2}\|\tilde{x} - \tilde{y}_n\|^2 + \gamma L_{\theta\delta}(\tilde{x}, \tilde{y}_n)\right\}. \tag{20}$$

2. Second step

$$\tilde{y}_{n+1} = \mathrm{Pr}_{\tilde{y}\in\tilde{U}\times\tilde{V}}\left\{\tilde{y}_n - \gamma\nabla_{\tilde{y}}L_{\theta\delta}\left(\tilde{x}, \bar{\tilde{y}}_n\right)\right\}. \tag{21}$$

Now we are ready to show the following theorem for the convergence to a unique saddle point of the iterative method given in Eq. (20) and Eq. (21).

Let us introduce the operator

$$\Phi\tilde{y} = \begin{pmatrix} \dfrac{\partial}{\partial\tilde{x}}L_{\theta\delta}(\tilde{x}, \tilde{y}) \\ \dfrac{\partial}{\partial\tilde{y}}L_{\theta\delta}(\tilde{x}, \tilde{y}) \end{pmatrix}$$

such that

$$(\Phi\tilde{y}^*, \tilde{y} - \tilde{y}^*) \geq 0, \quad \tilde{y} \in \tilde{Y} \tag{22}$$

Theorem 6. *Let us consider a Stacklebrg game for a set $\mathcal{N}$ of defenders $(l = \overline{1,n})$ and a set $\mathcal{M}$ of attackers $(r = \overline{1,m})$. Let $L_{\theta\delta}(\tilde{x}, \tilde{y})$ be differentiable in $\tilde{x}$ and $\tilde{y}$, whose partial derivative with respect to $\tilde{y}$ satisfies the Lipschitz condition with positive constant K such that $\|\tilde{y}_{n+1} - \tilde{y}_n\| \leq \gamma K\|\tilde{y}_n - \tilde{y}_n\|$. Let $\{\tilde{y}_n\}_{n\in\mathbb{N}}$ be a sequence defined by the local search algorithm given in Eq. (20) and Eq. (21) by*

$$\begin{aligned} \bar{\tilde{y}}_n &= \arg\min_{\tilde{x}\in\tilde{U}\times\tilde{V}}\left\{\tfrac{1}{2}\|\tilde{x} - \tilde{y}_n\|^2 + \gamma L_{\theta\delta}(\tilde{x}, \tilde{y}_n)\right\} \\ \tilde{y}_{n+1} &= \mathrm{Pr}_{\tilde{y}\in\tilde{U}\times\tilde{V}}\left\{\tilde{y}_n - \gamma\nabla_{\tilde{y}}L_{\theta\delta}\left(\tilde{x}, \bar{\tilde{y}}_n\right)\right\} \end{aligned} \tag{23}$$

then, the sequence $\{\tilde{y}_n\}_{n\in\mathbb{N}}$ converges to some Stackelberg equilibrium point $\tilde{y}^* \in \tilde{U}\times\tilde{V}$. In addition, for any $\alpha \in (0,1)$, $K^l \leq n\max_{l=1,n}K^l$ and $K^r \leq m\max_{r=1,m}K^r$ the stepsize parameter is given by

$$\gamma_n = \gamma_n(\alpha) < \sqrt{\frac{(1-\alpha)}{2}}\max\left\{\left[n\left(\max_{l=1,n}K^l\right)\right]^{-1}, \left[m\left(\max_{r=1,m}K^r\right)\right]^{-1}\right\}$$

such that $0 < \gamma_n \leq \gamma_0$.

Proof.

a) For any $\gamma > 0$ we have that $(\tilde{y}_n - \tilde{y}_{n+1}) + \gamma_n\partial L_{\theta\delta}(\tilde{x}_{n+1}, \tilde{y}_{n+1}) + \mathcal{C}_{\tilde{Y}}(\tilde{y}_{n+1}) \ni 0$ where $\mathcal{C}_{\tilde{Y}}(\tilde{y}_{n+1})$ is the normal cone of $\tilde{Y}$ at $\tilde{y}_{n+1}$. This is true if $\tilde{y}_n - \tilde{y}_{n+1} - \nabla L_{\theta\delta}(\tilde{x}_{n+1}, \tilde{y}_{n+1}) \in \mathcal{C}_{\tilde{Y}}(\tilde{y}_{n+1})$ when $\nabla_{\tilde{y}}L_{\theta\delta}(\tilde{x}_{n+1}, \tilde{y}_{n+1}) \in \partial L_{\theta\delta}(\tilde{x}_{n+1}, \tilde{y}_{n+1})$. Then, we have that

$$\tilde{y}^* = \mathrm{Pr}_{\tilde{y}\in\tilde{Y}}\{\tilde{y}^* - \gamma\nabla_{\tilde{y}}L_{\theta\delta}(\tilde{x}^*, \tilde{y}^*)\}$$

b) Let us rewrite the iterative method presented in Eq. (23) as follows

$$\left(\bar{\tilde{y}}_n - \tilde{y}_n + \gamma\Phi\tilde{y}_n, \tilde{y} - \tilde{y}_n\right) \geq 0 \tag{24}$$

$$\left(\tilde{y}_{n+1} - \tilde{y}_n + \gamma\Phi\bar{\tilde{y}}_n, \tilde{y} - \tilde{y}_{n+1}\right) \geq 0 \tag{25}$$

Summing Eq. (24) and Eq. (25) we obtain

$$\begin{aligned} &\left(\bar{\tilde{y}}_n - \tilde{y}_n, \tilde{y}_{n+1} - \tilde{y}x_n\right) + \gamma\left(\Phi\tilde{y}_n, \tilde{y}_{n+1} - \bar{\tilde{y}}_n\right) \\ &+ (\tilde{y}_{n+1} - \tilde{y}_n, \tilde{y}^* - \tilde{y}_{n+1}) + \gamma(\Phi\bar{\tilde{y}}_n, \tilde{y}^*_\delta - \tilde{y}_{n+1}) \geq 0 \end{aligned} \tag{26}$$

Developing Eq. (26) we have

$$\begin{aligned} &\frac{1}{2}\|\tilde{y}_{n+1} - \tilde{y}_n\|^2 - \frac{1}{2}\|\bar{\tilde{y}}_n - \tilde{y}_n\|^2 - \frac{1}{2}\|\tilde{y}_{n+1} - \bar{\tilde{y}}_n\|^2 + \gamma\left(\Phi\tilde{y}_n, \tilde{y}_{n+1} - \bar{\tilde{y}}_n\right) \\ &+ \frac{1}{2}\|\tilde{y}_n - \tilde{y}^*_\delta\|^2 - \frac{1}{2}\|\tilde{y}_{n+1} - \tilde{y}_n\|^2 - \frac{1}{2}\|\tilde{y}^*_\delta - \tilde{y}_{n+1}\|^2 + \gamma(\Phi\bar{\tilde{y}}_n, \tilde{y}^*_\delta - \tilde{y}_{n+1}) \geq 0 \end{aligned} \tag{27}$$

hence

$$\begin{aligned}&\left\|\bar{\tilde{y}}_n-\tilde{y}_n\right\|^2+\left\|\tilde{y}_{n+1}-\bar{\tilde{y}}_n\right\|^2+\|\tilde{y}_{n+1}-\tilde{y}_\delta^*\|^2-\|\tilde{y}_n-\tilde{y}_\delta^*\|^2\\&\le 2\gamma\left(\Phi\tilde{y}_n,\tilde{y}_{n+1}-\bar{\tilde{y}}_n\right)+2\gamma(\Phi\bar{\tilde{y}}_n,\tilde{y}_\delta^*-\bar{\tilde{y}}_n+\bar{\tilde{y}}_n+\tilde{y}_{n+1})\\&=2\gamma\left(\Phi\tilde{y}_n-\Phi\bar{\tilde{y}}_n,\tilde{y}_{n+1}-\bar{\tilde{y}}_n\right)+2\gamma(\Phi\bar{\tilde{y}}_n-\Phi\tilde{y}_\delta^*+M\tilde{y}_\delta^*,\tilde{y}_\delta^*-\bar{\tilde{y}}_n)\\&=2\gamma\left(\Phi\tilde{y}-\Phi\bar{\tilde{y}}_n,\tilde{y}_{n+1}-\bar{\tilde{y}}_n\right)+2\gamma(\Phi\bar{\tilde{y}}_n-\Phi\tilde{y}_\delta^*,\tilde{y}_\delta^*-\bar{\tilde{y}}_n)+2\gamma(M\tilde{y}_\delta^*,\tilde{y}_\delta^*-\bar{\tilde{y}}_n)\end{aligned}\tag{28}$$

given that $\|\tilde{y}_{n+1}-\tilde{y}_n\|^2=\|\tilde{y}_{n+1}-\mu\|^2+\|\mu-\tilde{y}_n\|^2+2(\tilde{y}_{n+1}-\mu,\mu-\tilde{y}_n)$ and $(\tilde{y}_{n+1}-\mu,\mu-\tilde{y}_n)=\frac{1}{2}\|\tilde{y}_{n+1}-\tilde{y}_n\|^2-\|\tilde{y}_{n+1}-\mu\|^2-\|\mu-\tilde{y}_n\|^2$.

If we let $\tilde{y}=\bar{\tilde{y}}^n$ from Eq. (22) we have $(\Phi\tilde{y}_\delta^*,\bar{\tilde{y}}_n-\tilde{y}_\delta^*)\ge 0$ if $(\Phi\tilde{y}_\delta^*,\tilde{y}_\delta^*-\bar{\tilde{y}}_n)\le 0$. Now, from the fact that for convex functions $f(w)\ge f(z)+(\nabla f(z),w-z)$ and for concave functions $f(w)\le f(z)+(\nabla f(z),w-z)$ it is easy to show that

$$\left(\Phi\bar{\tilde{y}}_n-\Phi\tilde{y}^*,\bar{\tilde{y}}^n-\tilde{y}^*\right)\ge 0\tag{29}$$

As a result, from Eq. (28) we obtain

$$\begin{aligned}&\left\|\tilde{y}_n-\bar{\tilde{y}}_n\right\|^2+\left\|\tilde{y}_{n+1}-\bar{\tilde{y}}_n\right\|^2+\|\tilde{y}_{n+1}-\tilde{y}_\delta^*\|^2\le\|\tilde{y}_n-\tilde{y}_\delta^*\|^2\\&+2\gamma\left(\Phi\tilde{y}_n-\Phi\bar{\tilde{y}}_n,\tilde{y}_{n+1}-\bar{\tilde{y}}_n\right)+2\gamma(\Phi\bar{\tilde{y}}_n-\Phi\tilde{y}_\delta^*,\tilde{y}_\delta^*-\bar{\tilde{y}}_n)\\&+2\gamma(\Phi\tilde{y}_\delta^*,\tilde{y}_\delta^*-\bar{\tilde{y}}_n)\le 2\gamma\left\|\Phi\tilde{y}_n-\Phi\bar{\tilde{y}}_n\right\|\left\|\tilde{y}_{n+1}-\bar{\tilde{y}}_n\right\|+\|\tilde{y}_n-\tilde{y}_\delta^*\|^2\\&+2\gamma(\Phi\bar{\tilde{y}}_n-\Phi\tilde{y}_\delta^*,\tilde{y}_\delta^*-\bar{\tilde{y}}_n)+2\gamma(\Phi\tilde{y}_\delta^*,\tilde{y}_\delta^*-\bar{\tilde{y}}_n)\le\|\tilde{y}_n-\tilde{y}_\delta^*\|^2\\&+2\gamma\left\|\Phi\tilde{y}_n-\Phi\bar{\tilde{y}}_n\right\|\left\|\tilde{y}_{n+1}-\bar{\tilde{y}}_n\right\|\le\|\tilde{y}_n-\tilde{y}_\delta^*\|^2+2\gamma K^+\left\|\tilde{y}_n-\bar{\tilde{y}}_n\right\|\left\|\tilde{y}_{n+1}-\bar{\tilde{y}}_n\right\|\\&\le\|\tilde{y}_n-\tilde{y}_\delta^*\|^2+2\gamma K^+\left\|\tilde{y}_n-\bar{\tilde{y}}_n\right\|^2+2\gamma K^+\left\|\tilde{y}_{n+1}-\bar{\tilde{y}}_n\right\|^2\end{aligned}$$

where $K^+=\max\left\{\left[n\left(\max_{l=1,n}K^l\right)\right]^{-1},\left[m\left(\max_{r=1,m}K^r\right)\right]^{-1}\right\}$. Then

$$\begin{aligned}\|\tilde{y}_{n+1}-\tilde{y}_\delta^*\|^2\le\|\tilde{y}_n-\tilde{y}_\delta^*\|^2+\left(2\gamma^2\left(K^+\right)^2-1\right)\left\|\tilde{y}_n-\bar{\tilde{y}}_n\right\|^2\\+\left(2\gamma^2\left(K^+\right)^2-1\right)\left\|\tilde{y}_{n+1}-\bar{\tilde{y}}_n\right\|^2\end{aligned}$$

If we take $2\gamma^2\left(K^+\right)^2-1\le-\alpha$ we have

$$\|\tilde{y}_{n+1}-\tilde{y}_\delta^*\|^2+\left\|\tilde{y}_n-\bar{\tilde{y}}_n\right\|^2+\left\|\tilde{y}_{n+1}-\bar{\tilde{y}}_n\right\|^2\le\|\tilde{y}_n-\tilde{y}_\delta^*\|^2,\quad n\in\mathbb{N}\tag{30}$$

implying that $\|\tilde{y}_n-\tilde{y}_\delta^*\|^2\le\|\tilde{y}_0-\tilde{y}_\delta^*\|^2$ for $n\in\mathbb{N}$.

Summing Eq. (30) we have that $\sum_{n=0}^{\infty}\left[\left\|\tilde{y}_n-\bar{\tilde{y}}_n\right\|^2+\left\|\tilde{y}_{n+1}-\bar{\tilde{y}}_n\right\|^2\right]$ converges. Then,

$$\left\|\tilde{y}_n-\bar{\tilde{y}}_n\right\|\to 0,\quad\left\|\tilde{y}_{n+1}-\bar{\tilde{y}}_n\right\|\to 0,\quad n\to\infty$$

As a result,

$$\|\tilde{y}_{n+1}-\tilde{y}_n\|\le\left\|\tilde{y}_{n+1}-\bar{\tilde{y}}_n\right\|+\left\|\bar{\tilde{y}}_n-\tilde{y}_n\right\|$$

and

$$\|\tilde{y}_{n+1}-\tilde{y}_n\|\to 0,\quad n\to\infty$$

Then, for the sequence $\{\tilde{y}_n\}$ there is a limit $\tilde{y}^*$ which is a Stackelberg equilibrium point. □

7. Planning

Let $(\Omega,\mathcal{F},\mathcal{P})$ be a probability space where Ω is a sample space; $\mathcal{F}$ is an -algebra of measurable subsets (events) of Ω; and $\mathcal{P}$ is a probability measure on $\mathcal{F}$ ($\mathcal{P}$ satisfies the following Kolmogorov axioms (Poznyak, 2008)).

The capture of an attacker is defined as follows:

Definition 7. *Let Ω be a probability space. The capture event $\omega\in\Omega$ is given when the defender and attacker are located at the same state at time n, i.e.*

$$\sum_{j=1}^{N}\chi\left(\omega:s^l(n)=s_j\wedge s^r(n)=s_j\right)=\sum_{j=1}^{N}\chi\left(\omega:s^l(n)=s_j\right)\chi\left(w:s^r(n)=s_j\right)$$

Remark 8. *The capture event of all the attackers is given by*

$$\sum_{l=1}^{n}\sum_{r=1}^{m}\sum_{j=1}^{N}\chi\left(\omega:s^l(n)=s_j\right)\chi\left(\omega:s^r(n)=s_j\right)\tag{31}$$

Let us consider that $P\{\omega:A\in\mathcal{F}\}=E\{\chi(\omega:A\in\mathcal{F})\}$, then the probabilities distribution vectors P^l $(l=\overline{1,n})$ of the defenders and P^r $(r=\overline{1,m})$ of the attackers of the Satackelberg security game are given by

$$P_{j,n}^l\left\{\omega:s_n^l=s_j\right\}=\sum_{i=1}^{N}p_{ij}^{l*}P_{j,n-1}^l\left\{\omega:s^l(n-1)=s_i\right\}$$

and

$$P_{j,n}^r\left\{\omega:s_n^r=s_j\right\}=\sum_{i=1}^{N}p_{ij}^{r*}P_{j,n-1}^r\left\{\omega:s^r(n-1)=s_i\right\}$$

where

$$p_{ij}^{l*}=\sum_{k=1}^{K}p_{jlik}^{l}d_{(k^*li)}^{l*}\tag{32}$$

and

$$p_{ij}^{r^*}=\sum_{k=1}^{K}p_{jlik}^{r}d_{(k^*li)}^{r^*}\tag{33}$$

Definition 9. *Let Ω be a probability space and let $\omega\in\Omega$ be a capture event The total probability P_n of converging to a state j at time n for all the defenders and attackers is given by*

$$\begin{aligned}&P_{j,n}=\sum_{l=1}^{n}\sum_{r=1}^{m}P_{j,n}^l\left\{\omega:s^l(n)=s_j\right\}\cdot\\&P_{j,n}^r\left\{\omega:s^r(n)=s_j\right\}=\sum_{l=1}^{n}\sum_{r=1}^{m}P_{j,n}^lP_{j,n}^r\end{aligned}\tag{34}$$

Let us introduce the ergodicity properties for the Satackelberg security game.

Theorem 10. (the ergodic theorem). *For some state $j_0\in(1,\dots,N)$ of a homogeneous (stationary) Markov chain with the transition matrix p and some $n>0$, $\varepsilon\in(0,1)$ for all $i\in\mathcal{G}$ let*

$$p_{ij_0}(n):=P\left(X_{(n)}=s_{(j_0)}\mid X_{(0)}=s_{(i)}\right)\ge\varepsilon.\tag{35}$$

Then for any initial-state distribution $P\{X_{(0)} = s_{(i)}\}$ *and for any* $i, j = 1, \ldots, N$ *there exists the limit*

$$P^*_{(j)} := \lim_{t\to\infty} p_{(ij)}(t) \tag{36}$$

such that for any $t \geq 0$ *this limit is reachable with an* ***exponential rate****, namely,*

$$|p_{(ij)}(t) - P^*_{(j)}| \leq (1-\varepsilon)^t = e^{-\alpha t} \tag{37}$$

where $\alpha := |\ln(1-\varepsilon)|$ and $[z]$ is the integer part of $z \in \mathbb{R}$.

Proof. See Clempner and Poznyak (2014) □

Lemma 11. (on a stationary state distribution). *Suppose that (34) holds. Then for any* $j = 1, \ldots, N$ *and for any*

$$P_j(n) = P_n := P\{X_t = s_{(j)}\}$$

the following property holds

$$|P_n - P^*| \leq (1-\delta)^t \tag{38}$$

where p^*_j *as in Eq. (36).*

Proof. See Clempner and Poznyak (2014) □

Theorem 12. *If for a finite Markov chain that is controllable by the fixed local-optimal policy, the lower bound estimate of the ergodicity coefficient*

$$\chi_{erg} := \min_{n_0} \max_{j=1,\ldots,N} \min_{i=1,\ldots,N} \tilde{p}^*_{ij}(n_0) \tag{39}$$

is strictly positive, that is, $\chi_{erg} > 0$*, then the following properties hold:*

1) *there exists a unique stationary distribution*

$$P^* = \lim_{n\to\infty} P_n$$

2) *The convergence of the current-state distribution to the stationary one is exponential:*

$$\begin{aligned} &|P_i(n) - P^*_i| \leq C\exp\{-Dn\} \\ &C = \frac{1}{1-\chi^l_{erg}}, \quad D = \frac{1}{n^*_0}\ln C, \\ &n^*_0 = \arg\min_{n_0}\left[\max_{j=1,\ldots,N}\min_{i=1,\ldots,N}\tilde{p}^*_{ij}(n_0)\right] \end{aligned} \tag{40}$$

Proof. See Clempner and Poznyak (2014) □

Then, the following theorem holds.

Theorem 13. *The capture condition given in Eq. (31) is satisfied if the convolution of the probabilities distribution vectors* P^l $(l = \overline{1,n})$ *and* P^r $(r = \overline{1,m})$ *of the Stackelberg security game converge in probability, i.e.*

$$\lim_{n\to\infty} P\left(\left|P^l_n P^r_n - P^l P^r\right| \geq \varepsilon\right) = 0, \quad \text{for } \varepsilon < 0, \quad l = \overline{1,n} \text{ and } r = \overline{1,m}$$

Proof. By ergodicity we know that

$$P^l_n \underset{n\to\infty}{\to} P^l \text{ and } P^r_n \underset{n\to\infty}{\to} P^r$$

converge in distribution. Now $\left|P^l_n P^r_n - P^l P^r\right| = \left|P^l_n P^r_n - P^l P^r_n + P^l P^r_n - P^l P^r\right|$. By the triangle inequality we have $\left|P^l_n P^r_n - P^l P^r_n + P^l P^r_n - P^l P^r\right| \leq \left|P^l_n P^r_n - P^l P^r_n\right| + \left|P^l P^r_n - P^l P^r\right|$. In fact, $\left|P^l_n P^r_n - P^l P^r_n\right| + \left|P^l P^r_n - P^l P^r\right| = |P^r_n||P^l_n - P^l| + |P^l||P^r_n - P^r|$. Then,

$$|P^r_n||P^l_n - P^l| + |P^l||P^r_n - P^r| \leq |\eta||P^l_n - P^l| + |\eta||P^r_n - P^r| \tag{41}$$

for $\eta = \max(P^l, P^r)$.

So, if $\left|P^l_n P^r_n - P^l P^r\right| \geq \varepsilon$ we have that $|P^l||P^l_n - P^l| \geq \varepsilon/2$ and $|P^r||P^r_n - P^r| \geq \varepsilon/2$. Hence, from Eq. (41) we have

$$\begin{aligned} \lim_{n\to\infty} P\left(\left|P^l_n P^r_n - P^l P^r\right| \geq \varepsilon\right) &= \lim_{n\to\infty} P\left(|\eta||P^l_n - P^l| \geq \varepsilon/2\right) \\ &\quad + \lim_{n\to\infty} P\left(|\eta||P^r_n - P^r| \geq \varepsilon/2\right) \\ &= \lim_{n\to\infty} P\left(|P^l_n - P^l| \geq \frac{\varepsilon}{\eta 2}\right) + \lim_{n\to\infty} P\left(|P^r_n - P^r| \geq \frac{\varepsilon}{\eta 2}\right) \\ &= 0 \end{aligned} \tag{42}$$

Taking the limit in Eq. (42) we have $P\left(\left|P^l_n P^r_n - P^l P^r\right| \geq \varepsilon\right) = P\left(|\eta||P^l_n - P^l| \geq \varepsilon/2\right) + P\left(|\eta||P^r_n - P^r| \geq \varepsilon/2\right)$

Then, it follows

$$P_{j,n} = \sum_{l=1}^{n}\sum_{r=1}^{m} P^l_{j,n}\{\omega : s^l(n) = s_j\} P^r_{j,n}\{\omega : s^r(n) = s_j\} \quad \square$$

Corollary 14. *The convolution of the probabilities distribution vectors* P^l $(l = \overline{1,n})$ *and* P^r $(r = \overline{1,m})$ *of the Stackelberg security game converge in probability then it converges in distribution.*

The computational algorithm for the patrol schedule is as follows:

Algorithm 1: Patrol schedule

1. Fix an initial state i for each player ι
2. **For** each player ι
3. **For** the matrix π^ι_{ik} select an action $a = a_k$ with random $k \in \{\overline{1,K}\}$ distributed according to the vector π^ι_{ik} for a fixed $i \in \{\overline{1,N}\}$.
4. **For** the matrix z^ι_{ik} and $k \in \{\overline{1,K}\}$ compute the entropy given by by $\hat{p}^\iota_i = -z^\iota_{ik}\ln z^\iota_{ik}$
5. **For** the matrix $p^\iota_{j|ik}$ compute $\hat{p}^\iota_{j|ik} = \lambda p^\iota_{j|ik} + (1-\lambda)\hat{p}^\iota_i$ with λ randomly selected
6. Normalize $\hat{p}^\iota_{j|ik}$ projecting the values into the simplex: $\Pr : \hat{p}^\iota_{j|ik} \to \mathcal{S}^N$
7. **For** the matrix $\hat{p}^\iota_{i\,j|k}$ find the state s_j with random $j \in \{\overline{1,N}\}$ distributed according to the stochastic vector $\hat{p}^\iota_{j|ik}$ for a fixed $i \in \{\overline{1,N}\}$ and action $a = a_k$ for $k \in \{\overline{1,K}\}$.
8. Add s_j to the patrol schedule and update the initial value of i with j.
9. Repeat steps (2) to (8) until the capture condition $\chi(\omega : s^l(n) = s_j)\chi(\omega : s^r(n) = s_j)$ is satisfied, $(l = \overline{1,n})$ and $(r = \overline{1,m})$

The Algorithm 1 is described as follows. First, establish an initial state i (which can be randomly selected). At each iteration time, execute the following steps for each player ι (the index ι involves defenders and attackers). We start the process finding the action $a = a_k$ to be executed in the patrol schedule using a random $k \in \{\overline{1,K}\}$ distributed according to the stochastic vector π^{ι}_{ik} for a given $i \in \{\overline{1,N}\}$. Given the state s_i and the action a_k, computes the vector $\hat{p}^{\iota}_{jik} = \lambda p^{\iota}_{jik} - (1-\lambda)\hat{p}^{\iota}_{i}$. We improve the information of the transition matrix p^{ι}_{jik} with additional information. The notion of entropy $(-z^{\iota}_{ik} \ln z^{\iota}_{ik})$ we use is classical in the field of information theory and, its interpretation corresponds to the order in which the targets can be attacked. Defenders may coordinate actions in order to gain better access to information and improve their limited security resources. Normalize $\hat{p}^{\iota}_{jik}$ projecting the values into the simplex employing the operator: $\Pr : \hat{p}^{\iota}_{jik} \to \mathcal{S}^{N}$.

Given $\hat{p}^{\iota}_{jik}$, the state s_i and the action a_k determines the next state s_j to be considered in the patrol schedule. This is done by using the information in the vector $\hat{p}^{\iota}_{jik}$ and selecting randomly $j \in \{\overline{1,N}\}$. Next, we add the state s_j, found in the previous step, to the patrol schedule, as well as update the initial variables of the algorithm. All of this completes a single iteration. We continue iterating until a stopping-time condition is accomplished, and then, the patrol schedule is completed.

8. Application example

This example represents a Stackelberg security Markov game consisting of two police officers (defenders) $l = 1, 2$ that try to stop two delinquents (attackers) $r = 3, 4$. Let us consider $N = 6$ and $K = 2$. The goal is to protect six control points (identified by black circled numbers in the map of the Fig. 1) selected using a rational choice model for prevention of common (minor) delinquency.

The transition matrices are as follows

$$p^{(1)}_{jli1} = \begin{bmatrix} 0.2564 & 0.7202 & 0.0005 & 0.0230 & 0.9487 & 0.7060 \\ 0.0115 & 0.3802 & 0.2628 & 0.3455 & 0.7577 & 0.0318 \\ 0.2662 & 0.0333 & 0.3895 & 0.3111 & 0.7431 & 0.2769 \\ 0.0513 & 0.5530 & 0.3145 & 0.0812 & 0.3922 & 0.0462 \\ 0.6324 & 0.1576 & 0.4218 & 0.8491 & 0.6555 & 0.0971 \\ 0.0975 & 0.9706 & 0.9157 & 0.9340 & 0.1712 & 0.8235 \end{bmatrix}$$

$$p^{(1)}_{jli2} = \begin{bmatrix} 0.0077 & 0.0217 & 0.0346 & 0.9360 & 0.7513 & 0.5472 \\ 0.0758 & 0.3909 & 0.3933 & 0.14014 & 0.2551 & 0.1386 \\ 0.2946 & 0.1857 & 0.1726 & 0.3470 & 0.5060 & 0.1493 \\ 0.0611 & 0.2346 & 0.5648 & 0.1395 & 0.6991 & 0.2575 \\ 0.4387 & 0.4456 & 0.6551 & 0.5853 & 0.8909 & 0.8407 \\ 0.3816 & 0.6463 & 0.1626 & 0.2238 & 0.9593 & 0.2543 \end{bmatrix}$$

$$p^{(2)}_{jli1} = \begin{bmatrix} 0.0239 & 0.8486 & 0.0471 & 0.0804 & 0.5688 & 0.3112 \\ 0.0239 & 0.8486 & 0.0471 & 0.0804 & 0.4694 & 0.5285 \\ 0.0975 & 0.1078 & 0.6152 & 0.1794 & 0.0119 & 0.1656 \\ 0.2028 & 0.2719 & 0.3546 & 0.1707 & 0.3371 & 0.6020 \\ 0.1966 & 0.5853 & 0.3804 & 0.9340 & 0.1622 & 0.2630 \\ 0.2511 & 0.5497 & 0.5678 & 0.1299 & 0.7943 & 0.6541 \end{bmatrix}$$

$$p^{(2)}_{jli2} = \begin{bmatrix} 0.3579 & 0.0507 & 0.2801 & 0.3113 & 0.1818 & 0.5499 \\ 0.5572 & 0.3785 & 0.0120 & 0.0523 & 0.2638 & 0.1450 \\ 0.1751 & 0.2076 & 0.4572 & 0.1602 & 0.1455 & 0.8530 \\ 0.0829 & 0.2980 & 0.3511 & 0.2681 & 0.1361 & 0.6221 \\ 0.2290 & 0.0782 & 0.8173 & 0.4314 & 0.8693 & 0.3510 \\ 0.9133 & 0.4427 & 0.8687 & 0.9106 & 0.5797 & 0.5132 \end{bmatrix}$$

$$p^{(3)}_{jli1} = \begin{bmatrix} 0.3288 & 0.2152 & 0.3739 & 0.0821 & 0.5752 & 0.0430 \\ 0.1308 & 0.3639 & 0.4474 & 0.0578 & 0.0598 & 0.1690 \\ 0.1394 & 0.1225 & 0.4843 & 0.2538 & 0.2348 & 0.6491 \\ 0.5846 & 0.1538 & 0.0937 & 0.1679 & 0.3532 & 0.7317 \\ 0.1839 & 0.4909 & 0.7803 & 0.9421 & 0.8212 & 0.6477 \\ 0.2400 & 0.4893 & 0.3897 & 0.9561 & 0.0154 & 0.4509 \end{bmatrix}$$

$$p^{(3)}_{jli2} = \begin{bmatrix} 0.4525 & 0.3901 & 0.1061 & 0.0513 & 0.3507 & 0.2077 \\ 0.2963 & 0.6256 & 0.4359 & 0.7948 & 0.9390 & 0.3012 \\ 0.7447 & 0.7802 & 0.4468 & 0.6443 & 0.8759 & 0.4709 \\ 0.1890 & 0.0811 & 0.3063 & 0.3786 & 0.5502 & 0.2305 \\ 0.6868 & 0.9294 & 0.5085 & 0.8116 & 0.6225 & 0.8443 \\ 0.1835 & 0.7757 & 0.5108 & 0.5328 & 0.5870 & 0.1948 \end{bmatrix}$$

$$p^{(4)}_{jli1} = \begin{bmatrix} 0.2259 & 0.4302 & 0.2581 & 0.2217 & 0.0855 & 0.4886 \\ 0.1707 & 0.1848 & 0.4087 & 0.1174 & 0.2625 & 0.5785 \\ 0.2277 & 0.9049 & 0.5949 & 0.2967 & 0.8010 & 0.2373 \\ 0.4357 & 0.9797 & 0.2622 & 0.3188 & 0.0292 & 0.4588 \\ 0.3111 & 0.4389 & 0.6028 & 0.4242 & 0.9289 & 0.9631 \\ 0.9234 & 0.1111 & 0.7112 & 0.5079 & 0.7303 & 0.5468 \end{bmatrix}$$

$$p^{(4)}_{jli2} = \begin{bmatrix} 0.5211 & 0.3674 & 0.0987 & 0.1068 & 0.8909 & 0.5000 \\ 0.2316 & 0.9880 & 0.2619 & 0.6538 & 0.3342 & 0.4799 \\ 0.4889 & 0.0377 & 0.3354 & 0.4942 & 0.6987 & 0.9047 \\ 0.6241 & 0.8852 & 0.6797 & 0.7791 & 0.1978 & 0.6099 \\ 0.6791 & 0.9133 & 0.1366 & 0.7150 & 0.0305 & 0.6177 \\ 0.3955 & 0.7962 & 0.7212 & 0.9037 & 0.7441 & 0.8594 \end{bmatrix}$$

and the individual cost matrices for each player are defined by

$$J^{(1)}_{jli1} = \begin{bmatrix} 23 & 2 & 87 & 11 & 85 & 85 \\ 19 & 96 & 90 & 18 & 100 & 88 \\ 17 & 3 & 19 & 75 & 1 & 75 \\ 15 & 98 & 67 & 5 & 55 & 12 \\ 21 & 30 & 95 & 8 & 87 & 51 \\ 96 & 53 & 98 & 49 & 91 & 17 \end{bmatrix}$$

$$J^{(1)}_{jli2} = \begin{bmatrix} 832 & 272 & 175 & 116 & 655 & 132 \\ 929 & 868 & 245 & 81 & 27 & 760 \\ 170 & 742 & 641 & 361 & 786 & 926 \\ 884 & 448 & 809 & 829 & 923 & 833 \\ 388 & 710 & 854 & 215 & 493 & 260 \\ 383 & 945 & 399 & 792 & 835 & 214 \end{bmatrix}$$

$$J^{(2)}_{jli1} = \begin{bmatrix} 82 & 28 & 96 & 80 & 68 & 71 \\ 91 & 55 & 49 & 96 & 76 & 4 \\ 13 & 96 & 81 & 66 & 75 & 28 \\ 92 & 97 & 15 & 4 & 40 & 5 \\ 64 & 16 & 43 & 85 & 66 & 10 \\ 10 & 98 & 92 & 94 & 18 & 83 \end{bmatrix}$$

$$J^{(2)}_{jli2} = \begin{bmatrix} 40 & 725 & 927 & 920 & 105 & 260 \\ 470 & 581 & 9 & 642 & 470 & 879 \\ 151 & 541 & 825 & 106 & 220 & 189 \\ 992 & 706 & 768 & 269 & 923 & 760 \\ 428 & 6 & 998 & 764 & 321 & 32 \\ 956 & 783 & 228 & 806 & 858 & 643 \end{bmatrix}$$

$$J^{(3)}_{jli1} = \begin{bmatrix} 96 & 30 & 34 & 97 & 29 & 19 \\ 9 & 6 & 46 & 63 & 71 & 5 \\ 80 & 51 & 74 & 14 & 54 & 89 \\ 60 & 77 & 51 & 62 & 20 & 84 \\ 92 & 29 & 39 & 39 & 69 & 12 \\ 11 & 23 & 91 & 100 & 6 & 42 \end{bmatrix}$$

$$J^{(3)}_{jli2} = \begin{bmatrix} 567 & 13 & 692 & 639 & 291 & 662 \\ 377 & 187 & 35 & 595 & 562 & 603 \\ 213 & 486 & 489 & 499 & 634 & 474 \\ 793 & 839 & 972 & 568 & 931 & 357 \\ 146 & 142 & 113 & 427 & 978 & 476 \\ 490 & 733 & 744 & 77 & 94 & 672 \end{bmatrix}$$

$$J^{(4)}_{jli1} = \begin{bmatrix} 13 & 21 & 17 & 84 & 35 & 10 \\ 58 & 67 & 24 & 76 & 55 & 47 \\ 95 & 98 & 6 & 63 & 47 & 59 \\ 26 & 63 & 91 & 40 & 65 & 19 \\ 99 & 7 & 80 & 36 & 52 & 62 \\ 35 & 31 & 38 & 9 & 82 & 6 \end{bmatrix}$$

$$J^{(4)}_{jli2} = \begin{bmatrix} 58 & 76 & 85 & 49 & 27 & 66 \\ 85 & 80 & 67 & 7 & 66 & 39 \\ 50 & 30 & 97 & 90 & 14 & 31 \\ 44 & 12 & 95 & 50 & 64 & 35 \\ 15 & 38 & 12 & 78 & 39 & 92 \\ 3 & 83 & 65 & 7 & 77 & 46 \end{bmatrix}$$

The resulting strategies for the defenders and attackers are the following

$$\pi^{1^*}_{kli} = \begin{bmatrix} 0.3292 & 0.6708 \\ 0.0753 & 0.9247 \\ 0.1789 & 0.8211 \\ 0.4644 & 0.5356 \\ 0.4454 & 0.5546 \\ 0.2439 & 0.7561 \end{bmatrix} \quad \pi^{2^*}_{kli} = \begin{bmatrix} 0.3586 & 0.6414 \\ 0.1720 & 0.8280 \\ 0.2769 & 0.7231 \\ 0.0754 & 0.9246 \\ 0.4737 & 0.5263 \\ 0.3154 & 0.6846 \end{bmatrix}$$

$$\pi^{3^*}_{kli} = \begin{bmatrix} 0.3292 & 0.6708 \\ 0.0753 & 0.9247 \\ 0.1789 & 0.8211 \\ 0.4644 & 0.5356 \\ 0.4454 & 0.5546 \\ 0.2439 & 0.7561 \end{bmatrix} \quad \pi^{4^*}_{kli} = \begin{bmatrix} 0.3586 & 0.6414 \\ 0.1720 & 0.8280 \\ 0.2769 & 0.7231 \\ 0.0754 & 0.9246 \\ 0.4737 & 0.5263 \\ 0.3154 & 0.6846 \end{bmatrix}$$

The Fig. 2 and Fig. 3 show the convergence of the strategies of defenders and, Fig. 4 and Fig. 5 show the convergence of the strategies of attackers. The convergence of the parameter ξ is presented in the Fig. 6.

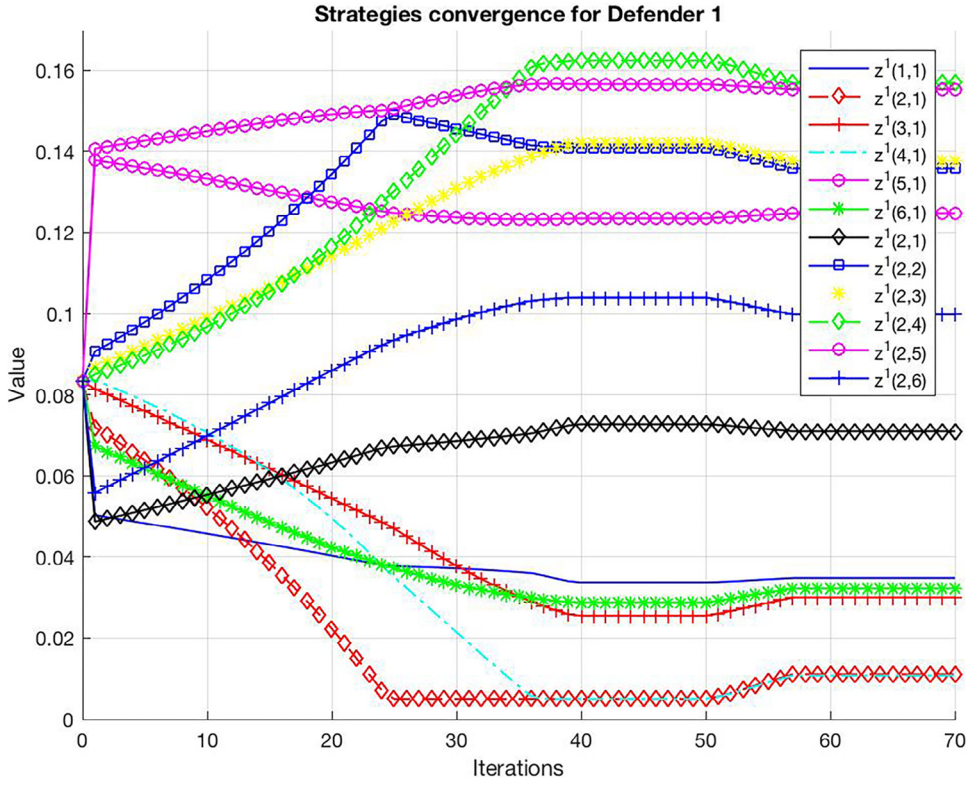


Fig. 2 – Convergence of the strategies of the defender 1.

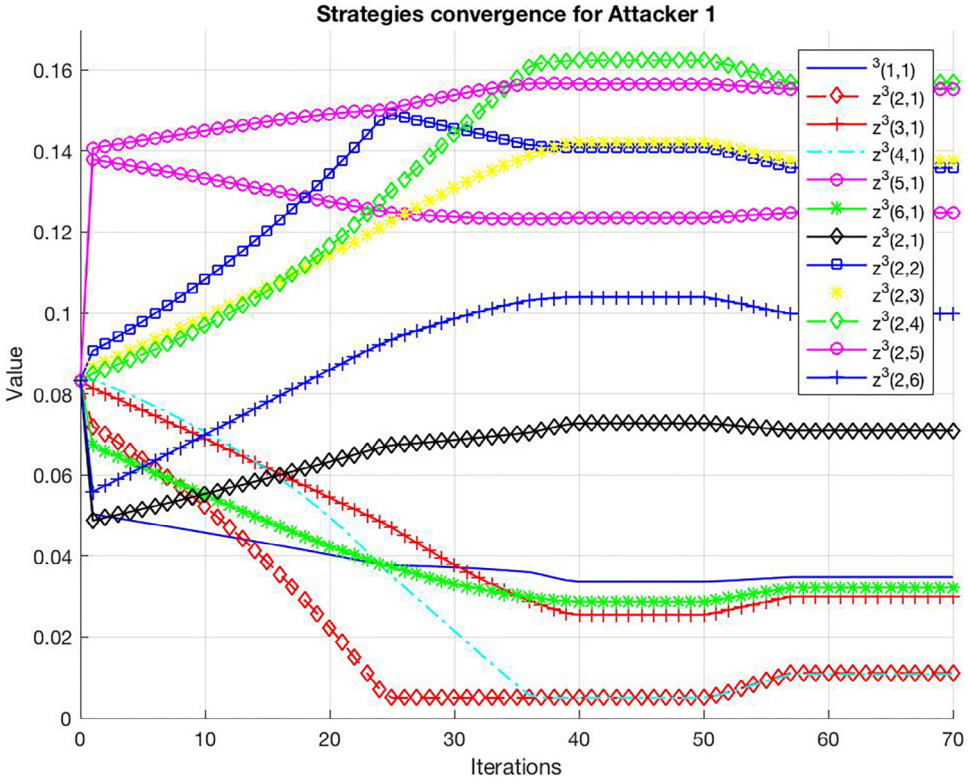


Fig. 3 – Convergence of the strategies of the defender 2.

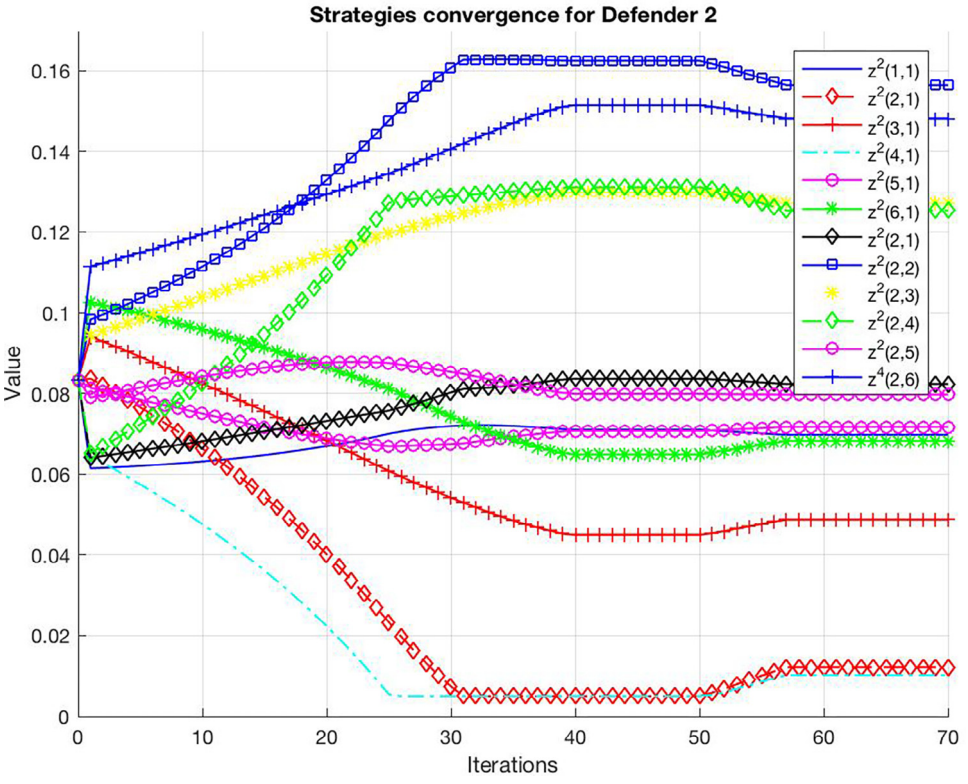


Fig. 4 – Convergence of the strategies of the attacker 1.

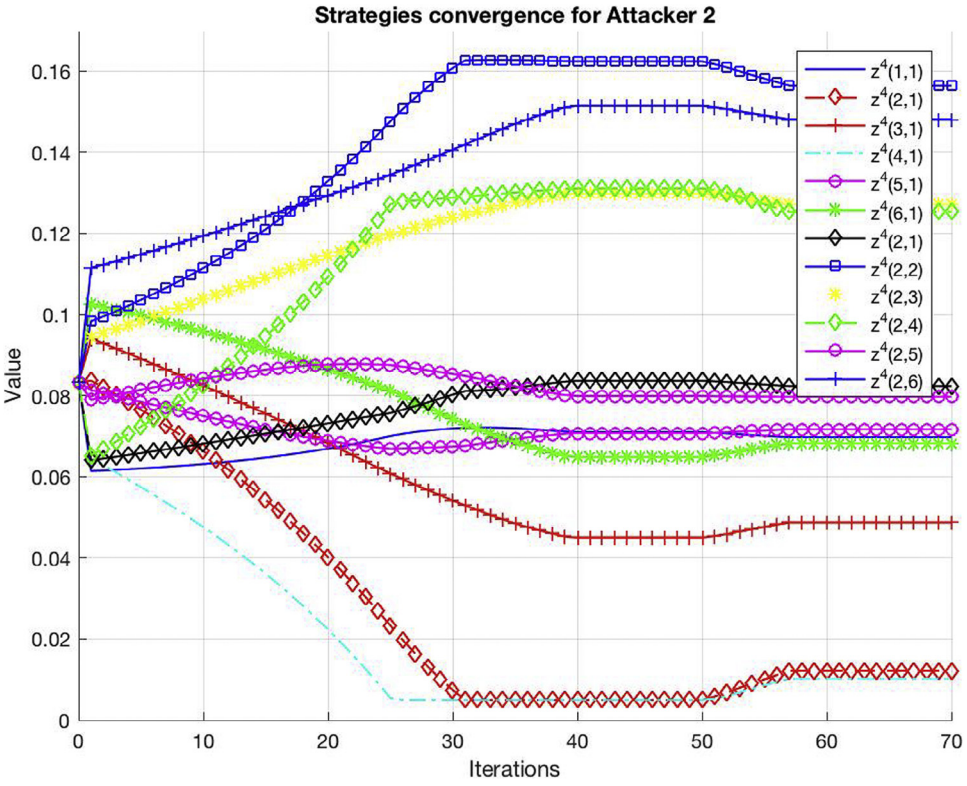


Fig. 5 – Convergence of the strategies of the attacker 2.

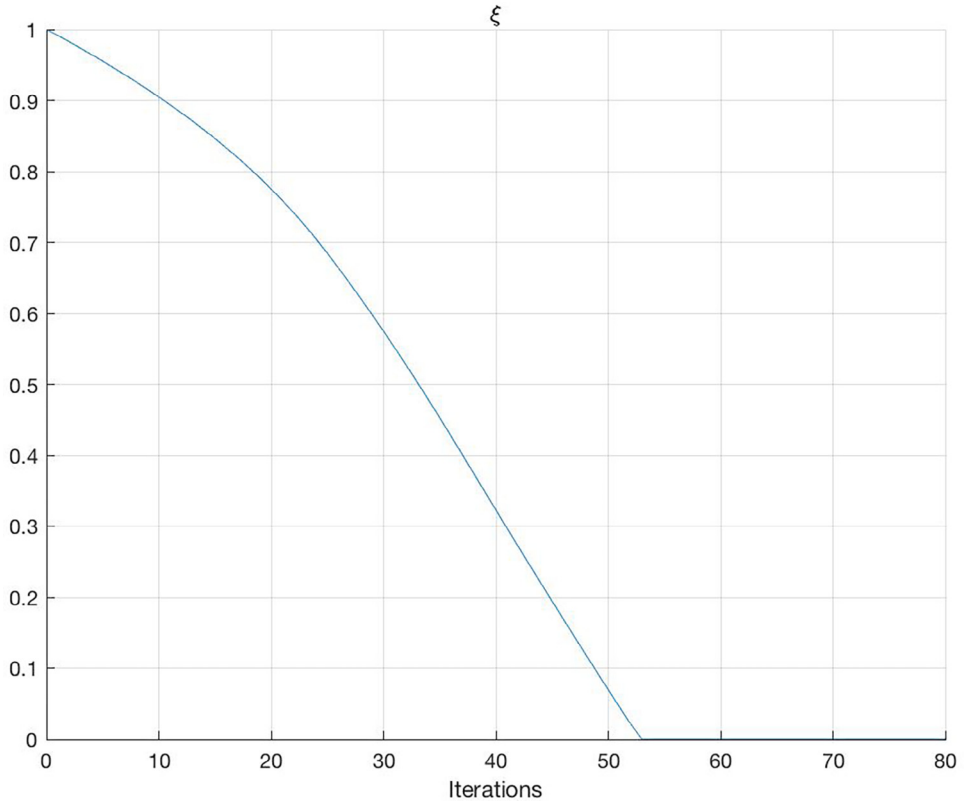


Fig. 6 – Convergence of the strategies of the parameter ξ.

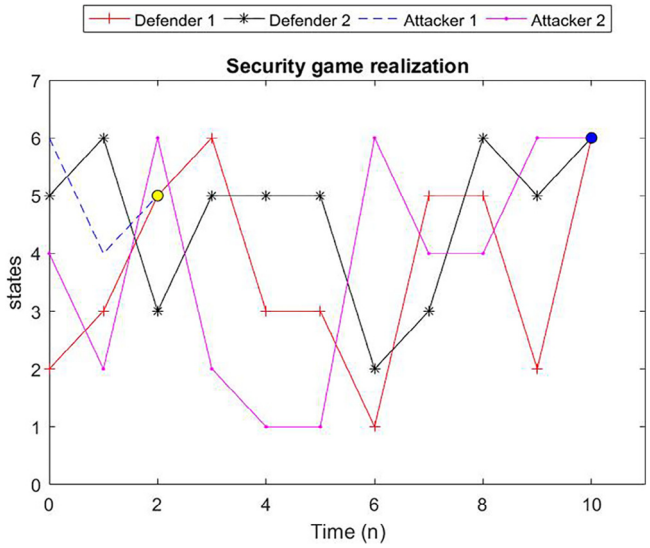


Fig. 7 – Patrol realization 1.

Now, we consider the random walk process presented in Algorithm 1. A realization of the game is shown in Fig. 7 where the attacker 1 is caught at state 5 after 2 steps by the defender 1, and attacker 2 is caught at state 6 after 10 steps by the defender 2, so the game is over. A different realization of the game is shown in in Fig. 8. Here, we have that the attacker 1 is caught at state 6 after 3 steps by cooperation of the defender 1 and defender 2, as well as we have that the attacker 2 is caught at state 2 after 8 steps, again by defender 2, and then the game is over.

9. Conclusions

Security is a fundamental problem around the world, wherever the challenge is protecting people and critical infrastructure. The dynamics of the game is given by defenders that protect a set of targets against attackers. This framework is represented using Stackelberg security Markov games, where defenders commit first to a randomized strategy to which the attackers devise the best response after surveillance of this strategy. There are many real-world security domains in which defenders may cooperate in order to more effectively capture attackers: metro, ports, airports and air transportation, etc (Tambe, 2011). Defenders may coordinate actions in order to gain better access to information and improve their limited security resources. This coordination can result in higher levels of coverage at all times.

In order to solve this problem, this paper presents a new solution approach for Stackelberg security Markov games. The game involves defenders engaged in a cooperative game and attackers are related non-cooperatively in a Nash game restricted by a Stackelberg game. For solving the collusion of the defenders we employed the Nash bargaining approach. For computing the Stackelberg equilibrium point we proposed a new approach based in an iterative method conceptualized in two half-steps. We transformed the problem in a system of equations where each one is an independent optimization problem. We showed the convergence and the rate of convergence of the method. A random walk solution for planning the patrol schedule is presented. The random walk solution involves the notion of entropy and its interpretation corresponds to the order in which the targets will be attacked. The usefulness of the suggested approach is illustrated by a numerical example.

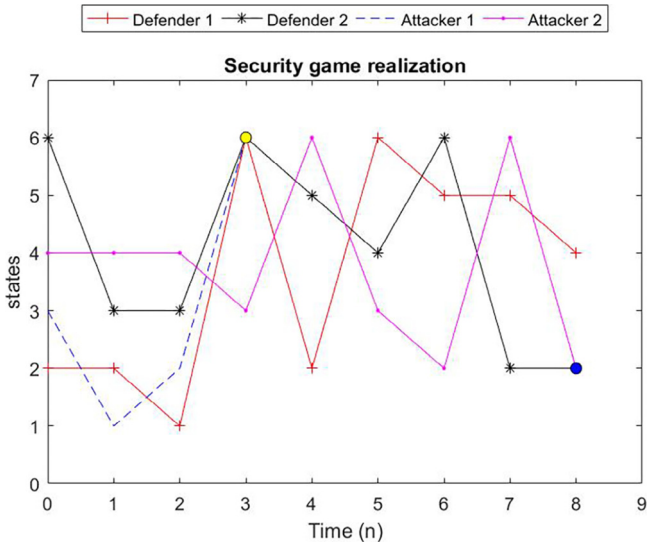


Fig. 8 – Patrol realization 2.

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